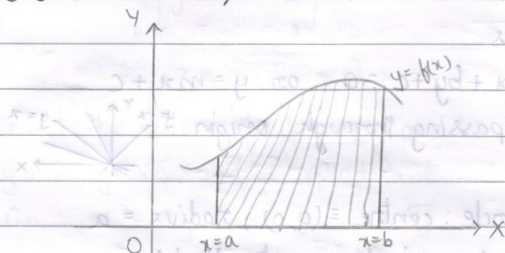


APPLICATION OF INTEGRALS

Area enclosed between the curve $y = f(x)$, x -axis and the ordinates $x = a$, $x = b$.



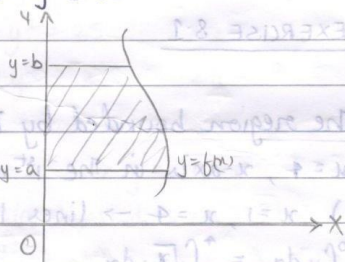
$$A = \int_a^b f(x) dx$$

$dx \rightarrow$ w.r.t 'x'

i.e., Area measured over x -axis

$x = a$, $x = b$ } lines $\parallel$ to y -axis

Area enclosed between the curve $y = f(x)$, The lines $y = a$, $y = b$ and y -axis.

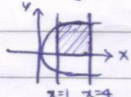


$$A = \int_a^b x \cdot dy$$

$dy \rightarrow$ w.r.t 'y' $\Rightarrow$ Area measured over y -axis

$x = f(y)$

$y = a$, $y = b$ } lines $\parallel$ to x -axis

General Lines and Curves:1. $x = k \rightarrow$ line $\parallel^e$ to y -axis $x = 0 \rightarrow$ y -axis $y = k \rightarrow$ line $\parallel^e$ to x -axis $y = 0 \rightarrow$ x -axisGeneral line: $ax + by + c = 0$ or $y = mx + c$ ** $y = mx \rightarrow$ lines passing through origin $x^2 + y^2 = a^2 \rightarrow$ circle: centre $\equiv (0,0)$, radius $= a$ $(x-h)^2 + (y-k)^2 = a^2 \rightarrow$ circle: centre $\equiv (h,k)$ $y^2 = 4ax$ } Parabola $x^2 = 4ay$ }
 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ } Ellipses $y = \frac{b}{a} \sqrt{a^2 - x^2}$
 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, b > a$
EXERCISE 8.11.] Find the area of the region bounded by the curve $y^2 = x$ and $x = 1, x = 4, x$ -axis in the 1st quadrant.Gn: $y^2 = x$ (parabola) $x = 1, x = 4 \rightarrow$ lines $\parallel^e$ to y -axis

$$\text{Area} = \int_a^b y \, dx = \int_1^4 \sqrt{x} \, dx$$

$$= \left[\frac{2}{3} x\sqrt{x} \right]_1^4$$

$$= \frac{2}{3} [8 - 1] = \frac{14}{3} \text{ sq. unit}$$

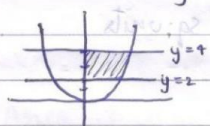
- 2) Find the area of the region bounded by $y^2 = 9x$, $x=2$, $x=4$, x -axis in 1st quadrant.

Gn: $y^2 = 9x$, $y = \sqrt{9x} = 3\sqrt{x}$

$$A = \int_a^b y \cdot dx = \int_2^4 3\sqrt{x} \, dx = \left[\frac{3 \cdot \frac{2}{3} x\sqrt{x}}{1} \right]_2^4 = 2(8 - 2\sqrt{2}) \text{ sq. units}$$

- 3) Find the area of the region bounded by $x^2 = 4y$, $y=2$, $y=4$, y -axis and 1st quadrant.

Gn: $x^2 = 4y \Rightarrow x = 2\sqrt{y}$



$$A = \int_a^b x \cdot dy = \int_2^4 2\sqrt{y} \cdot dy$$

$$A = \left[2 \cdot \frac{2}{3} y\sqrt{y} \right]_2^4$$

$$= \frac{4}{3} [8 - 2\sqrt{2}] \text{ sq. units}$$

Eg 1] Find the area enclosed by the circles

(a) $x^2 + y^2 = a^2$ πa^2

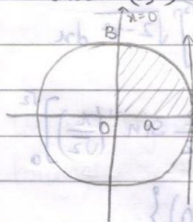
(b) $x^2 + y^2 = 8$ 8π $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$

(c) $x^2 + y^2 = 25$ 25π $x^2 + y^2 = a^2 \Rightarrow y = \sqrt{a^2 - x^2}$

(d) $x^2 + y^2 = 12$ 12π

a) $x^2 + y^2 = a^2 \Rightarrow y = \sqrt{a^2 - x^2}$

center = (0,0), radius = a



from fig:

Area of circle = 4 x Area of $\widehat{OAB}$ (sector)

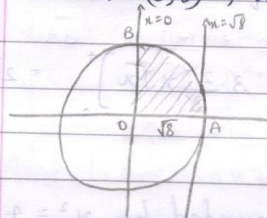
$$A = 4 \int_0^a y \, dx = 4 \int_0^a \sqrt{a^2 - x^2} \, dx$$

$$A = 4 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) \right]_0^a$$

$$= 4 \left\{ 0 + \frac{a^2}{2} \left(\frac{\pi}{2} - 0 \right) \right\}$$

$$A = 4 \cdot \frac{\pi a^2}{2} = 2\pi a^2 \text{ sq. units}$$

b) $x^2 + y^2 = 8 \Rightarrow y = \sqrt{8 - x^2}$
 Center $= (0, 0)$, radius $= \sqrt{8}$



Area of circle $= 4 \times$ Area of $\triangle OAB$

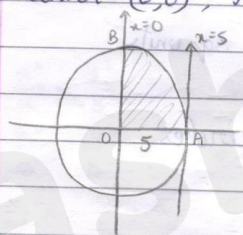
$$A = 4 \int_0^{\sqrt{8}} y \cdot dx = 4 \int_0^{\sqrt{8}} \sqrt{8 - x^2} dx$$

$$A = 4 \left[\frac{x}{2} \sqrt{8 - x^2} + \frac{8}{2} \sin^{-1} \left(\frac{x}{\sqrt{8}} \right) \right]_0^{\sqrt{8}}$$

$$= 4 \left\{ 0 - \frac{8}{2} \left(\frac{\pi}{2} - 0 \right) \right\}$$

$$= 4 \cdot \frac{8\pi}{2} = 8\pi \text{ sq. units}$$

c) $x^2 + y^2 = 25 \Rightarrow y = \sqrt{25 - x^2}$
 center $= (0, 0)$, radius $= 5$



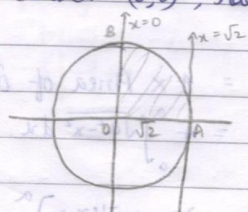
$$A = 4 \int_0^5 y \cdot dx = 4 \int_0^5 \sqrt{25 - x^2} dx$$

$$A = 4 \left[\frac{x}{2} \sqrt{25 - x^2} + \frac{25}{2} \sin^{-1} \left(\frac{x}{5} \right) \right]_0^5$$

$$= 4 \left\{ 0 - \frac{25}{2} \left(\frac{\pi}{2} - 0 \right) \right\}$$

$$= 4 \cdot \frac{25\pi}{2} = 25\pi \text{ sq. units}$$

d) $x^2 + y^2 = 2 \Rightarrow y = \sqrt{2 - x^2}$
 center $= (0, 0)$, radius $= \sqrt{2}$



$$A = 4 \int_0^{\sqrt{2}} y \cdot dx = 4 \int_0^{\sqrt{2}} \sqrt{2 - x^2} dx$$

$$A = 4 \left[\frac{x}{2} \sqrt{2 - x^2} + \frac{2}{2} \sin^{-1} \left(\frac{x}{\sqrt{2}} \right) \right]_0^{\sqrt{2}}$$

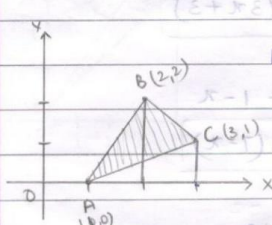
$$= 4 \left\{ 0 - \frac{2}{2} \left(\frac{\pi}{2} - 0 \right) \right\}$$

$$= 4 \cdot \frac{2\pi}{2} = 2\pi \text{ sq. units}$$

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**** Eg 9]** Find the area of the region bounded by the Δ^o whose vertices are $(1,0)$, $(2,2)$, $(3,1)$.

Sol] Gn: $A = (1,0)$, $B = (2,2)$, $C = (3,1)$



Eq. of sides: $\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$

$\overline{AB} : \frac{y-0}{2-0} = \frac{x-1}{2-1} \Rightarrow y = 2(x-1)$

$\overline{BC} : \frac{y-2}{1-2} = \frac{x-2}{3-2} \Rightarrow y-2 = 2-x \Rightarrow y = 4-x$

$\overline{AC} : \frac{y-0}{1-0} = \frac{x-1}{3-1} \Rightarrow y = \frac{1}{2}(x-1)$

Area of $\Delta ABC = \int_1^2 \text{Area under } \overline{AB} + \int_2^3 \text{Area under } \overline{BC} - \int_1^3 \text{Area under } \overline{AC} dx$

$$A = \int_1^2 2(x-1) dx + \int_2^3 (4-x) dx - \int_1^3 \frac{1}{2}(x-1) dx$$

$$= \left[\frac{2x^2}{2} - 2x \right]_1^2 + \left[4x - \frac{x^2}{2} \right]_2^3 - \frac{1}{2} \left[\frac{x^2}{2} - x \right]_1^3$$

$$= 3 - 2(1) + 4(1) - \frac{1}{2}(5) - \frac{1}{4}(8) + \frac{1}{2}(2)$$

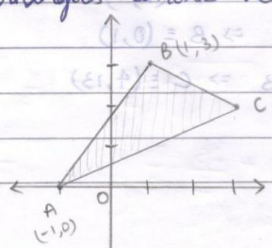
$$= 3 - 2 + 4 - 2.5 = 4 - \frac{5}{2} = \frac{3}{2} \text{ sq. units}$$

Verification: An of $\Delta = \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 1 \\ 3 & 1 & 1 \end{vmatrix} = \frac{1}{2} [1 + 1(-4)] = \frac{3}{2}$

EXERCISE 8.2

4] Using integration find the area of region bounded by the triangles whose vertices are $(-1,0)$, $(1,3)$, $(3,2)$.

Sol] Gn: $A = (-1,0)$, $B = (1,3)$, $C = (3,2)$



$$\text{Eq. of sides : } \frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\overline{AB} : \frac{y-0}{3-0} = \frac{x+1}{1+1} \Rightarrow y = \frac{1}{2}(3x+3)$$

$$\overline{BC} : \frac{y-3}{2-3} = \frac{x-1}{3-1} \Rightarrow 2y-6 = 1-x$$

$$y = \frac{1}{2}(7-x)$$

$$\overline{AC} : \frac{y-0}{2-0} = \frac{x+1}{3+1} \Rightarrow 4y = 2x+2$$

$$y = \frac{1}{2}(x+1)$$

$$\text{Area of } \triangle ABC = \int_{-1}^1 \frac{3}{2}(x+1) dx + \int_1^3 \frac{1}{2}(7-x) dx - \int_{-1}^3 \frac{1}{2}(x+1) dx$$

$$A = \frac{3}{2} \left[\frac{x^2}{2} + x \right]_{-1}^1 + \frac{1}{2} \left[7x - \frac{x^2}{2} \right]_1^3 - \frac{1}{2} \left[\frac{x^2}{2} + x \right]_{-1}^3$$

$$= \frac{3}{2}(2) + \frac{1}{2}(10) - \frac{1}{2}(8)$$

$$= 3+5-4 = 4 \text{ sq. units}$$

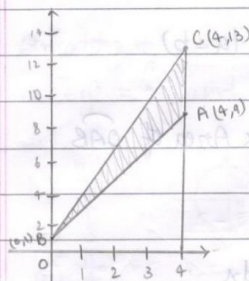
$$\begin{vmatrix} 1 & -1 & 0 & 1 \\ 2 & 1 & 3 & 1 \\ 3 & 2 & 1 & 1 \end{vmatrix} = \frac{1}{2}(-1+0+1(2-9))$$

$$= \frac{1}{2}(-8) = -4$$

3] Find the area of the region bounded by the curves $y = x^2 + 2$, $y = x$, $x = 0$ and $x = 3$.

5] Find the area of the triangular region whose sides have the equations $y = 2x+1$, $y = 3x+1$, $x = 4$.

Sol] Gn: $\overline{AB}: y = 2x+1$, $\overline{BC}: y = 3x+1$, $\overline{AC}: x = 4$ line ||^{ve} to y-axis
 pt. A: $\overline{AB} \& \overline{AC}: y = 2(4)+1 = 9 \Rightarrow A \equiv (4, 9)$
 pt. B: $\overline{AB} \& \overline{BC}: 0 = x \Rightarrow y = 1 \Rightarrow B \equiv (0, 1)$
 pt. C: $\overline{AC} \& \overline{BC}: y = 3(4)+1 \Rightarrow y = 13 \Rightarrow C \equiv (4, 13)$



$$\text{Area of } \triangle ABC = \int_0^4 \text{Area under } \overline{BC} - \int_0^4 \text{Area under } \overline{BA}$$

$$A = \int_0^4 (3x+1) dx - \int_0^4 (2x+1) dx$$

$$= \int_0^4 (3x+1) - (2x+1) dx$$

$$= \int_0^4 x \cdot dx = \left[\frac{x^2}{2} \right]_0^4 = \frac{16}{2} = 8 \text{ sq. units}$$

EXERCISE 8.1

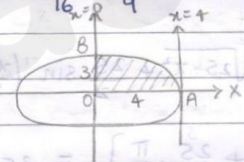
4] Find the area of the region bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$

Sol] Area, $A = \pi ab = \pi(4)(3) = 12\pi$ sq. units ($a > b$)

5] Find the area of region bounded by the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$

Sol] $A = \pi ab = \pi(2)(3) = 6\pi$ sq. units ($a < b$)

4] Gn: $\frac{x^2}{16} + \frac{y^2}{9} = 1$; $y = \frac{3}{4} \sqrt{16-x^2}$



From fig,

Area of ellipse = 4 x Area of OAB

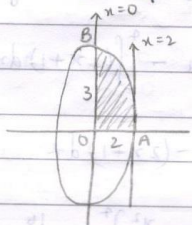
$$A = 4 \int_0^4 y \cdot dx$$

$$A = 4 \int_0^4 \frac{3}{4} \sqrt{16-x^2} dx$$

$$= 4 \cdot \frac{3}{4} \left[\frac{x}{2} \sqrt{16-x^2} + \frac{16}{2} \sin^{-1} \left(\frac{x}{4} \right) \right]_0^4$$

$$= 3 \left[8 \cdot \frac{\pi}{2} \right] = 12\pi \text{ sq. units}$$

5) Gn: $\frac{x^2}{4} + \frac{y^2}{9} = 1$; $y = \frac{3}{2} \sqrt{4-x^2}$ ($a < b$)



Area of ellipse = 4 x Area of $\widehat{OAB}$

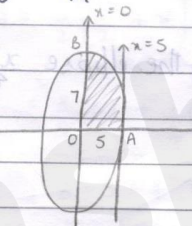
$$A = 4 \int_a^b y \, dx$$

$$A = 4 \int_0^2 \frac{3}{2} \sqrt{4-x^2} \, dx$$

$$A = 4 \cdot \frac{3}{2} \left[\frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1}\left(\frac{x}{2}\right) \right]_0^2$$

$$A = 6 \left[0 + \frac{4}{2} \cdot \frac{\pi}{2} \right] = 6\pi \text{ sq. units}$$

★] $\frac{x^2}{25} + \frac{y^2}{49} = 1$; $y = \frac{7}{5} \sqrt{25-x^2}$ ($a < b$)



Area of ellipse = 4 x Area of $\widehat{OAB}$

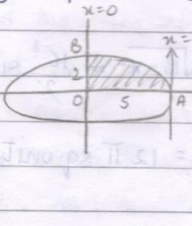
$$A = 4 \int_a^b y \, dx$$

$$A = 4 \int_0^5 \frac{7}{5} \sqrt{25-x^2} \, dx$$

$$A = 4 \cdot \frac{7}{5} \left[\frac{x}{2} \sqrt{25-x^2} + \frac{25}{2} \sin^{-1}\left(\frac{x}{5}\right) \right]_0^5$$

$$= 4 \cdot \frac{7}{5} \left[0 + \frac{25}{2} \cdot \frac{\pi}{2} \right] = 35\pi \text{ sq. units}$$

★] $\frac{x^2}{25} + \frac{y^2}{4} = 1$; $y = \frac{2}{5} \sqrt{25-x^2}$ ($a > b$)



Area of ellipse = 4 $\int_a^b y \, dx$

$$A = 4 \int_0^5 \frac{2}{5} \sqrt{25-x^2} \, dx$$

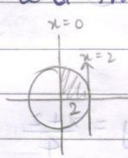
$$= 4 \cdot \frac{2}{5} \left[\frac{x}{2} \sqrt{25-x^2} + \frac{25}{2} \sin^{-1}\left(\frac{x}{5}\right) \right]_0^5$$

$$= 4 \cdot \frac{2}{5} \left[0 + \frac{25}{2} \cdot \frac{\pi}{2} \right] = 10\pi \text{ sq. units}$$

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12] Area lying in the first quadrant and bounded by the circle $x^2 + y^2 = 4$ and the lines $x=0$ and $x=2$ is

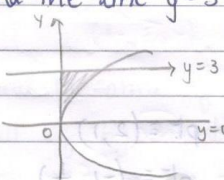
a) π
b) $\frac{\pi}{2}$
c) $\frac{\pi}{3}$
d) $\frac{\pi}{4}$



$x^2 + y^2 = 4$; center $(0,0)$; radius $= 2$
Area of req. region $= \frac{1}{4}$ Area of circle
 $A = \frac{1}{4} \pi (4)$
 $A = \pi$ sq. units

13] Area of the region bounded by the curve $y^2 = 4x$, y-axis and the line $y=3$ is

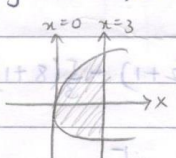
Sol.]



$y^2 = 4x \rightarrow$ parabola
 $y=3$, line $\parallel$ to x-axis
 $x = \frac{y^2}{4}$
limits: $y=0, y=3$
 $A = \int_0^3 x \cdot dy = \int_0^3 \frac{y^2}{4} dy = \frac{1}{4} \left[\frac{y^3}{3} \right]_0^3 = \frac{1}{4} \cdot \frac{1}{3} \cdot 27 = \frac{9}{4}$ sq. units

14] Find the area of the region bounded by the curve $y^2 = 4x$, line $x=3$

Sol.]



$y^2 = 4x \rightarrow$ parabola
 $y = \sqrt{4x} = 2\sqrt{x}$
 $x=3$, line $\parallel$ to y-axis
limits: $x=0, x=3$
 $A = \int_0^3 y \cdot dx = \int_0^3 2\sqrt{x} dx = 2 \cdot \frac{2}{3} x^{3/2} \Big|_0^3 = \frac{4}{3} \cdot 3\sqrt{3} = 4\sqrt{3}$ sq. units

10] Find the area bounded by the curve $x^2 = 4y$ and line $x = 4y - 2$

Sol] $x^2 = 4y \rightarrow (1)$ $x = 4y - 2 \rightarrow (2)$

$$(4y - 2)^2 = 4y \quad y = \frac{x^2}{4}$$

$$16y^2 + 4 - 16y - 4y = 0 \quad (2) \Rightarrow y = \frac{x+2}{4}$$

$$16y^2 - 20y + 4 = 0$$

$$4(4y^2 - 5y + 1) = 0$$

$$4y^2 - 4y - y + 1 = 0$$

$$4y(y-1) - 1(y-1) = 0$$

$$(4y-1)(y-1) = 0$$

$$y = 1, \frac{1}{4}$$

$y = 1 \Rightarrow x = 4(1) - 2 \Rightarrow x = 2$ pt $\equiv (2, 1)$

$y = \frac{1}{4} \Rightarrow x = 4(\frac{1}{4}) - 2 \Rightarrow x = -1$ pt $\equiv (-1, \frac{1}{4})$

Required area = $\widehat{OAB}$ = Area under line - Area under parabola

$$A = \int_{-1}^2 \left(\frac{x+2}{4} \right) dx - \int_{-1}^2 \frac{x^2}{4} dx$$

$$A = \int_{-1}^2 \frac{1}{4} (x+2-x^2) dx$$

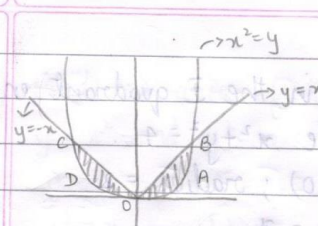
$$A = \frac{1}{4} \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \frac{1}{4} \left[\frac{3}{2} + 2(2+1) - \frac{1}{3}(8+1) \right]$$

$$A = \frac{1}{4} \left(\frac{3}{2} + 6 - 3 \right) = \frac{1}{4} \left(\frac{9}{2} \right) = \frac{9}{8} \text{ sq. units}$$

9] Find the area of the region bounded by the parabola $y = x^2$ and $y = |x|$

Sol] $x^2 = y \rightarrow (1)$; $y = |x| = \begin{cases} x & ; x \geq 0 \\ -x & ; x < 0 \end{cases}$

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Req. area = $\widehat{OABCD} = 2 \times \text{Area of } \triangle OAB$

$$A = \int_0^1 y \cdot dx = \int_0^1 (x - x^2) dx$$

pt. of intersection:

$$y = x, y = x^2$$

$$x^2 - x = 0$$

$$x(x-1) = 0$$

$$x = 0, 1$$

pt. $\equiv (0,0), (1,1)$

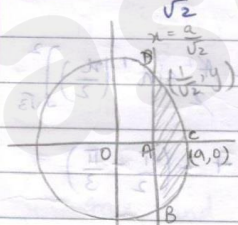
$$= 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$= 2 \left[\frac{1}{2} - \frac{1}{3} \right] = \frac{2}{6} \text{ sq. units} = \frac{1}{3} \text{ sq. units}$$

7] Find the area of the smaller part of the circle $x^2 + y^2 = a^2$ cut off by the line $x = \frac{a}{\sqrt{2}}$

Sol] $x^2 + y^2 = a^2 \rightarrow \text{①}$; center $\equiv (0,0)$; radius $= a$

line: $x = \frac{a}{\sqrt{2}}$ - line $\parallel$ to y-axis



① $\Rightarrow y = \sqrt{a^2 - x^2}$

Req. area = $\widehat{ABCD} = 2 \times \text{Area } \triangle ACD$

$$A = 2 \int_{\frac{a}{\sqrt{2}}}^a y \cdot dx$$

$$= 2 \int_{\frac{a}{\sqrt{2}}}^a \sqrt{a^2 - x^2} \cdot dx$$

$$= 2 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_{\frac{a}{\sqrt{2}}}^a$$

$$A = 2 \left[\frac{1}{2} \left(0 - \frac{a}{\sqrt{2}} \cdot \frac{a}{\sqrt{2}} \right) + \frac{a^2}{2} \left(\frac{\pi}{2} - \frac{\pi}{4} \right) \right]$$

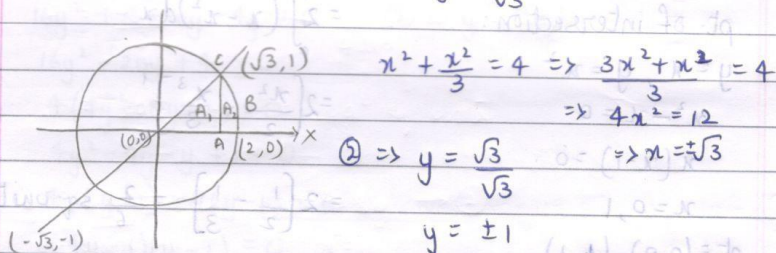
$$= 2 \left[-\frac{a^2}{4} + \frac{a^2}{2} \left(\frac{\pi}{2} - \frac{\pi}{4} \right) \right] = -\frac{a^2}{2} + a^2 \left(\frac{\pi}{2} - \frac{\pi}{4} \right)$$

$$A = \frac{a^2}{2} \left(\frac{\pi}{2} - 1 \right) \text{ sq. units}$$

6] Find the area of the region in the I quadrant enclosed by x-axis, $x = \sqrt{3}y$ and circle $x^2 + y^2 = 4$.

Sol) $x^2 + y^2 = 4 \rightarrow ①$; centre (0,0); radius = 2

line: $x = \sqrt{3}y \rightarrow ② \quad ② \Rightarrow y = \frac{x}{\sqrt{3}}$



$$① \Rightarrow y = \sqrt{4-x^2}$$

Req. Area = $\widehat{OBC} = \triangle OAC + \text{Area } \widehat{ABC}$

$$A = \int_a^b y \, dx = \int_0^{\sqrt{3}} \text{Area under line} + \int_{\sqrt{3}}^2 \text{Area under circle}$$

$$A = \int_0^{\sqrt{3}} \frac{x}{\sqrt{3}} \, dx + \int_{\sqrt{3}}^2 \sqrt{4-x^2} \, dx$$

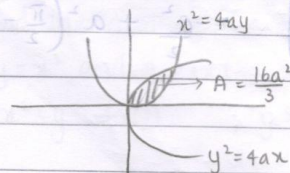
$$= \left[\frac{x^2}{2\sqrt{3}} \right]_0^{\sqrt{3}} + \left[\frac{x}{2} \sqrt{4-x^2} + \frac{x^2}{2} \sin^{-1} \left(\frac{x}{2} \right) \right]_{\sqrt{3}}^2$$

$$= \frac{1}{2\sqrt{3}} (3) + \frac{1}{2} (0 - \sqrt{3}) + 2 \left(\frac{\pi}{2} - \frac{\pi}{3} \right)$$

$$= \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} + 2 \cdot \frac{\pi}{6} = \frac{\pi}{3} \text{ sq. units}$$

✱✱

Area enclosed between 2 parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16a^2}{3}$ sq. units



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*** Eg6] Find the area of the region bounded by the parabola $y = x^2$ and $y^2 = x$

Sol] $y = x^2 \rightarrow (1)$
 $y^2 = x \rightarrow (2)$

pt. of intersection $x^2 = y$
 $x^4 = y^2$
 $x^4 = x$
 $x^4 - x = 0$
 $x(x^3 - 1) = 0 \Rightarrow x = 0, x = 1$
 $(1) \Rightarrow y = 0, 1$
pt. $(0,0), (1,1)$

Req. area = $\widehat{OABC} = \int_0^1 \text{Area under } (2) - \int_0^1 \text{Area under } (1) dx$

$$A = \int_0^1 \sqrt{x} - x^2 dx = \left[\frac{2}{3} x\sqrt{x} - \frac{x^3}{3} \right]_0^1$$

$$A = \frac{2}{3}(1) - \frac{1}{3} = \frac{1}{3} \text{ sq. units}$$

Shortcut : $4a = 1 \Rightarrow a = \frac{1}{4} \quad A = \frac{16a^2}{3} = \frac{16}{3} \left(\frac{1}{16} \right) = \frac{1}{3} \text{ sq. units}$

*] $y^2 = 4ax$ and $x^2 = 4ay \rightarrow (1)$
 $\Rightarrow y = 2\sqrt{ax} \quad \Rightarrow x = 2\sqrt{ay}$
 $(1) \Rightarrow x^2 = 4ay$
 $x^4 = 4^2 a^2 y^2$
 $x^4 = 4^2 a^2 (4ax)$
 $x(x^3 - 4^3 a^3) = 0$
 $x = 0, 4a$
 $(1) \Rightarrow y = 0, 4a$
pts are $(0,0), (4a, 4a)$

Req. area = $\triangle ABC = \int_0^{4a} \text{Area under (1)} - \text{Area under (2)} dx$

$$A = \int_0^{4a} \left(2\sqrt{a}\sqrt{x} - \frac{x^2}{4a} \right) dx = 2\sqrt{a} \cdot \frac{2}{3} x\sqrt{x} - \frac{1}{4a} \cdot \frac{x^3}{3} \Big|_0^{4a}$$

$$A = \frac{4\sqrt{a}}{3} a(2\sqrt{a}) - \frac{1}{4a(3)} (4a)(16a^2)$$

$$= \frac{16a^2}{3} (2-1) = \frac{16a^2}{3} \text{ sq. units}$$

*] $y^2 = 8x \rightarrow (1) \quad x^2 = 8y \rightarrow (2)$
 $y = \sqrt{8x} \quad x = \frac{y^2}{8}$

For intersection points:
 $x^4 = 8^2 y^2 = 8^2 (8x) \Rightarrow x^4 - 8^3 x = 0$
 $x(x^3 - 8^3) = 0$
 $x = 0, 2 \quad y = 0, 2$
 pts are $(0,0), (2,2)$

Req. area = $\triangle ABC = \int_0^2 \text{Area under (1)} - \text{Area under (2)} dx$

$$A = \int_0^2 \left(\sqrt{8}\sqrt{x} - \frac{x^2}{8} \right) dx = \sqrt{8} \cdot \frac{2}{3} x\sqrt{x} - \frac{1}{8} \cdot \frac{x^3}{3} \Big|_0^2$$

$$A = 2\sqrt{2} \cdot \frac{2}{3} (2-0)\sqrt{2} - \frac{1}{8} \left(\frac{8-0}{3} \right)$$

$$A = \frac{16}{3} - \frac{1}{3} = \frac{15}{3} = 5 \text{ sq. units}$$

*] $y^2 = 16x \rightarrow (1) \quad x^2 = 16y \rightarrow (2)$
 $x^4 = 16^2 y^2$
 $x^4 = 16^2 (16x) \Rightarrow x(x^3 - 16^3) = 0$
 $x = 0, 16$
 $\Rightarrow y = 0, 16$
 pts are $(0,0), (16,16)$

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Req. area = OABC = $\int_a^b y \, dx = \int_0^{16} y \, dx$

① $\Rightarrow y = 4\sqrt{x}$, ② $\Rightarrow y = \frac{x^2}{16}$

Area = $\int_0^{16} \text{Area under ①} - \text{Area under ②} \, dx$

$$= \int_0^{16} 4\sqrt{x} - \frac{x^2}{16} \, dx = \left[4 \cdot \frac{2}{3} x\sqrt{x} - \frac{x^3}{16 \cdot 3} \right]_0^{16}$$

$$= \frac{8}{3} (16\sqrt{16}) - \frac{16^3 - 0}{3 \cdot 16} = \frac{8}{3} (16 \times 4) - \frac{16^2 \cdot 16}{3 \cdot 16}$$

$$= \frac{8}{3} (64) - \frac{16^2}{3} = \frac{16}{3} (32 - 16) = \frac{256}{3} \text{ sq. units}$$

Verification : $y^2 = 16x$, $x^2 = 16y$

$4a = 16 \Rightarrow a = 4$

$A = \frac{16a^2}{3} = \frac{16(16)}{3} = \frac{256}{3}$

EXERCISE 8.2

Eg 10] Find the area of region enclosed b/w 2 circles
 $x^2 + y^2 = 4$ and $(x-2)^2 + y^2 = 4$.

Sol] $x^2 + y^2 = 4 \rightarrow$ ① $(x-2)^2 + y^2 = 4 \rightarrow$ ②

center $\equiv (0,0)$ $C \equiv (2,0)$, $r = 2$

radius = 2

$y = \sqrt{4-x^2}$ $y = \sqrt{4-(x-2)^2}$

$$\Rightarrow 4-x^2 = 4-(x-2)^2$$

$$x^2 = (x-2)^2$$

$$x^2 = x^2 + 4 - 4x$$

$$4x = 4$$

$$\boxed{x=1}$$

① $\Rightarrow y = \pm\sqrt{3}$

$\therefore$ pts are $(1, \sqrt{3})$, $(1, -\sqrt{3})$

Req: area = $\widehat{OABC} = 4 \times \widehat{BDC}$

$$A = 4 \int_a^b y \, dx = 4 \int_1^2 \sqrt{4-x^2} \, dx$$

$$A = 4 \left[\frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1} \left(\frac{x}{2} \right) \right]_1^2$$

$$= 4 \left\{ \frac{1}{2} (0 - \sqrt{3}) + 2 \left(\frac{\pi}{2} - \frac{\pi}{6} \right) \right\}$$

$$= -2\sqrt{3} + 8 \frac{\pi}{3} = 2 \left(\frac{4\pi}{3} - \sqrt{3} \right) \text{ sq. units}$$

2] Find the area bounded by the curves $(x-1)^2 + y^2 = 1$ and $x^2 + y^2 = 1$.

Sol] $x^2 + y^2 = 1 \rightarrow \textcircled{1} \quad (x-1)^2 + y^2 = 1$

$C \equiv (0,0), r=1$

$C \equiv (1,0), r=1$

$y = \sqrt{1-x^2}$

$y = \sqrt{1-(x-1)^2}$

$\Rightarrow 1-x^2 = 1-(x-1)^2$

$x^2 = x^2 + 1 - 2x$

$2x = 1 \Rightarrow x = \frac{1}{2}$

$\left[x = \frac{1}{2} \right] \quad \textcircled{1} \Rightarrow y = \pm \frac{\sqrt{3}}{2}$

pts are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right)$

Req: Area = $\widehat{OABC} = 4 \times \widehat{BDC}$

$$A = 4 \int_a^b y \, dx = 4 \int_{\frac{1}{2}}^1 \sqrt{1-x^2} \, dx$$

$$= 4 \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} \left(\frac{x}{1} \right) \right]_{\frac{1}{2}}^1$$

$$= 4 \left\{ \left(\frac{1}{2} \cdot \frac{\sqrt{3}}{2} \right) + \left(\frac{\pi}{2} - \frac{\pi}{6} \right) \frac{1}{2} \right\}$$

$$= \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

$= \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \text{ sq. units}$

